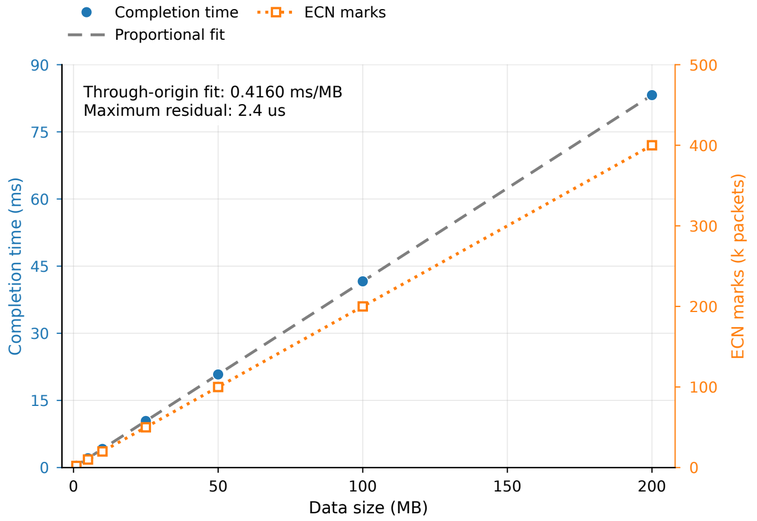
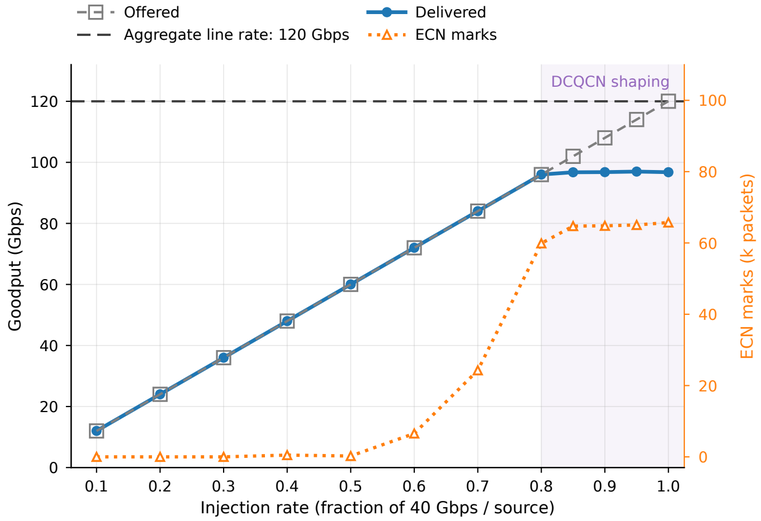
# ProtoBridge sweep summary (3-node tor, 40 Gbps RoCE, DCQCN)

ns-3.47 ProtoBridge

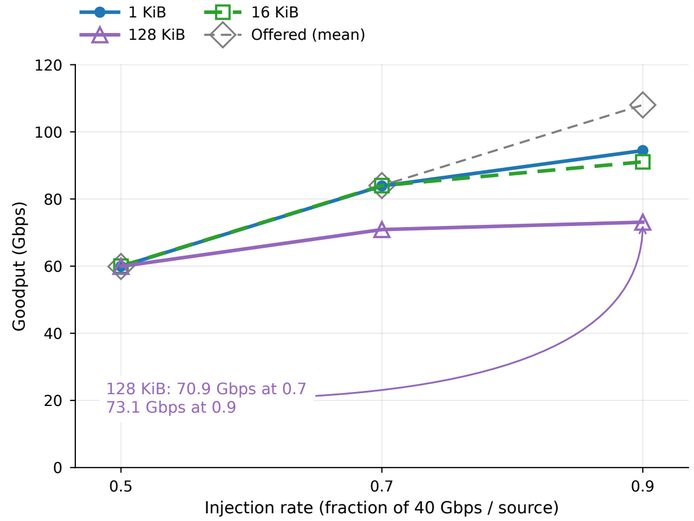
## (a) Data size: linear completion



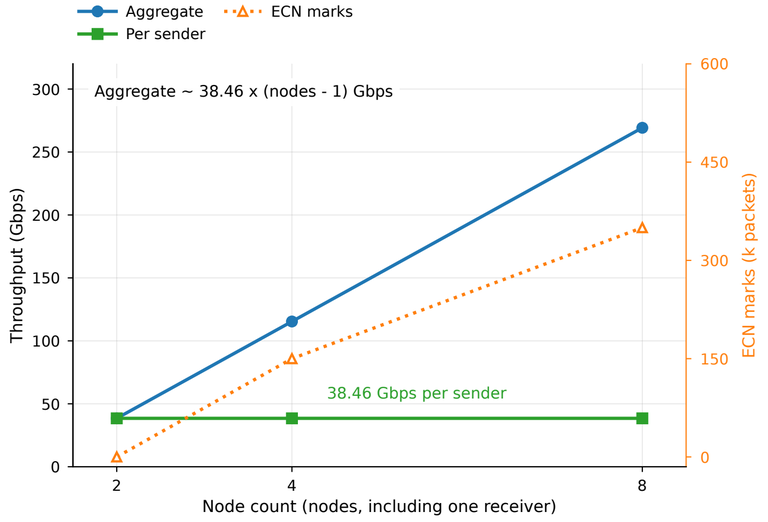

## (b) Offered vs. delivered traffic



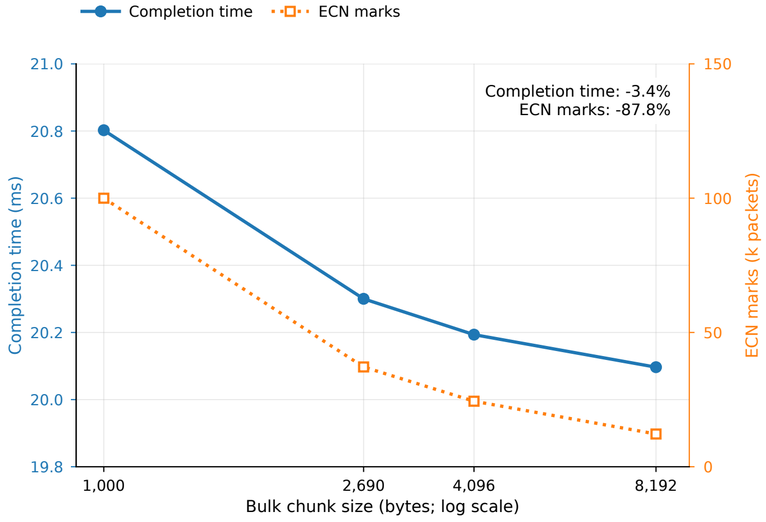

## (c) Large messages reach the knee sooner



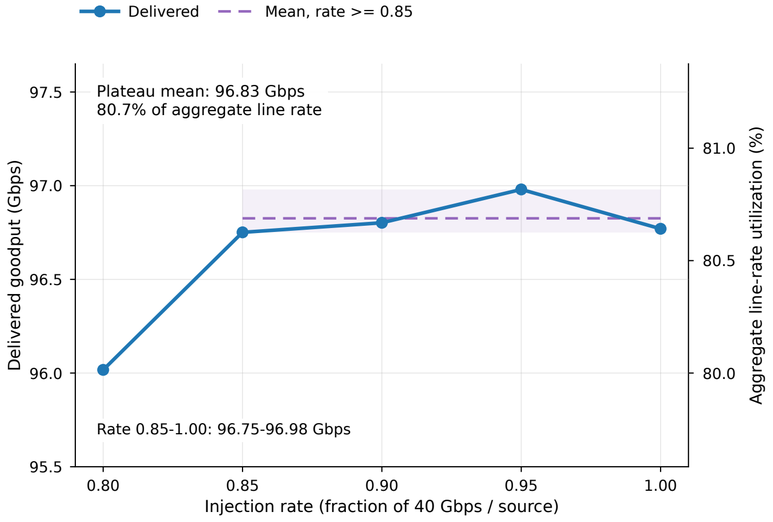

## (d) Incast throughput scales with senders




## (e) Larger chunks reduce time and ECN




## (f) Saturation plateau, magnified




ns-3.47 | Default: 3 nodes, single-leaf tor, 40 Gbps/link, 1 us latency. Node sweep varies node count. Incast panels use DCQCN.

Random traffic: uniform cross-node messages via SoHeader/ECN/DCQCN bulk path. MB = 1,000,000 bytes; KiB = 1,024 bytes; k packets = 1,000 packets. Chunk sweep: 50 MB.